

		direction
2(a)	From the given expression $p = \frac{1}{3} \rho \langle c^2 \rangle$, we have $p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle \text{ ----- (1) (where } m \text{ is the mass of 1 molecule)}$ Using the ideal gas equation $pV = NkT$, we have $p = \frac{NkT}{V} \text{ ----- (2)}$	

No.	Solution	Remarks
	Equating (1) and (2), we have $\frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle = \frac{NkT}{V}$ $\frac{1}{3} m \langle c^2 \rangle = kT$ $\frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT$ Since k is the Boltzmann constant, the average kinetic energy $\frac{1}{2} m \langle c^2 \rangle$ of a molecule of mass m, is directly proportional to the thermodynamics temperature T. (shown)	[1] for equating (1) and (2) [1] for correct derivation of expression [1] for the correct conclusion
2(b)(i)	Total pressure, $p_{\text{total}} = h\rho g + p_{\text{atm}}$ $= (15)(1.03 \times 10^3)(9.81) + 1.01 \times 10^5$ $= 2.526 \times 10^5$ $\approx 2.53 \times 10^5 \text{ Pa}$	[1] for correct method and numerical substitution [1] for correct answer
2(b)(ii)	Using $pV = nRT$, $(p_2 V_2)_{\text{water}} = (p_1 V_1)_{\text{tank}}$ $V_2 = \frac{p_1}{p_2} V_1$ $= \frac{2.32 \times 10^7}{2.526 \times 10^5} \times 9.4 \times 10^3$ $= 8.633 \times 10^5$ $\approx 8.63 \times 10^5 \text{ cm}^3$	[1] for correct expression and numerical substitution [1] for correct answer
2(c)	Pressure at depth of 35 m, $p_2 = h\rho g + p_{\text{atm}}$ $= (35)(1.03 \times 10^3)(9.81) + 1.01 \times 10^5$ $= 4.547 \times 10^5 \text{ Pa}$ Volume of air at depth of 35 m and temperature of 19 °C, $\left(\frac{p_2 V_2}{T_2} \right)_{35\text{m}} = \left(\frac{p_1 V_1}{T_1} \right)_{15\text{m}}$ $V_2 = \frac{p_1}{p_2} \times \frac{T_2}{T_1} \times V_1$ $= \frac{2.526 \times 10^5}{4.547 \times 10^5} \times \frac{19 + 273.15}{24 + 273.15} \times 8.633 \times 10^5$ $= 4.715 \times 10^5 \text{ cm}^3$	[1] for correct method and numerical substitution [1] for correct volume

No.	Solution	Remarks
	Duration of time the air will last for the diver $t = \frac{4.715 \times 10^5}{8.633 \times 10^5} \times 45$ $= 24.58$ $\approx 24.6 \text{ min}$	[1] for correct answer
3(c)	If path difference of the sound waves along XY is odd integer of	[1]